

$$a^2 - b^2 = (a+b)(a-b)$$

$$\frac{m^2 - n^2}{(m+n)^2} \times \frac{m^2 + mn + n^2}{m^3 - n^3}$$

$$= \frac{(m+n)(\cancel{m-n})}{(\cancel{m+n})(m+n)} \times \frac{\cancel{m^2 + mn + n^2}}{(\cancel{m-n})(\cancel{m^2 + mn + n^2})}$$

$$= \frac{1}{m+n}$$

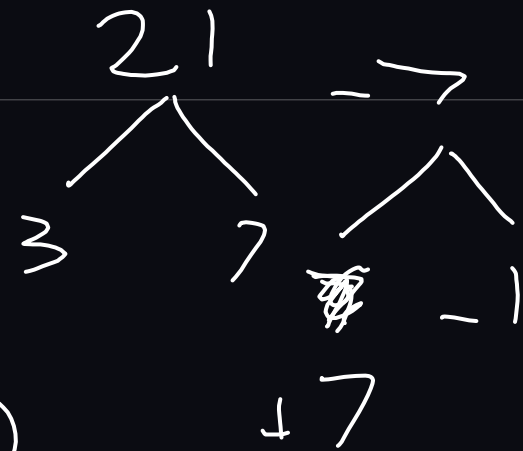
$$\frac{a^2+10a+21}{a^2+6a-7} \times \frac{a^2-1}{a+3}$$

$$= \frac{a^2+3a+7a+21}{a^2+7a-a-7} \times \frac{(a+1)(a-1)}{(a+3)}$$

$$= \frac{a(a+3)+7(a+3)}{a(a+7)-1(a+7)} \times \frac{(a+1)(a-1)}{(a+3)}$$

$$= \frac{\cancel{(a+3)}\cancel{(a+7)}}{\cancel{(a+7)}\cancel{(a-1)}} \times \frac{(a+1)\cancel{(a-1)}}{\cancel{(a+3)}}$$

$$= (a+1)$$



$$9y^2 - 12y + 2$$

$$\frac{12}{2} \text{ (6)}$$

$$9y^2 - 12y + 36 - 36 + 2$$

$$\rightarrow (3y + 6)^2 - 34 = 0$$

$$6^2 = \boxed{36}$$

$$(3y + 6)^2 = 34$$

$$3y + 6 = \pm \sqrt{34}$$

$$3y + 6 = \sqrt{34}$$

$$3y = \sqrt{34} - 6$$

$$y = \frac{\sqrt{34} - 6}{3}$$

OR

$$3y + 6 = -\sqrt{34}$$

$$3y = -\sqrt{34} - 6$$

$$y = \frac{-\sqrt{34} - 6}{3}$$

$$n=3$$

$$t_n = 600$$

$$n=7$$

$$t_n = 700$$

$$a + 6d = 700$$

$$a + 2d = 600$$

$$\begin{aligned} a &= ? \\ d &= ? \end{aligned} \quad \text{li)}$$

$$a + 9d = \square \quad A_n = 8$$

$$\begin{aligned} a &= 2 \\ d &= 2 \\ n &= ? \end{aligned}$$

$$\frac{n}{2} (2a + (n-1)d)$$

$$\begin{aligned} S_n &= \frac{(2a + (n-1)d) \cdot n}{2} \\ &= [4 + 3 \times 2] \times 2 \\ &= [10 \times 2] = [20] \end{aligned} \quad \frac{4 \times 2}{2}$$

$$2, 4, 6, 8, \dots, 16$$

$$\begin{aligned} t_n \quad A_n &= \frac{a + (n-1)d}{2} \\ 8 &= \frac{2 + (n-1)2}{2} \end{aligned}$$

$$8 - 2 = 2n - 2$$

$$\begin{aligned} 8 - 2 &= 2n \\ 6 &= 2n \\ 8 &= 2n \end{aligned}$$

$$n = 4$$